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AquaFusion AI: Project Proposal

Autonomous Aquaponics Management with Non-Feeder Plant-Based Biomass Cycling, Adaptive Overhead Spectrum Lighting, and Computer Vision Ecosystem Health Monitoring

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September 4, 2026

Abstract

Aquaponics presents an eco-friendly food production system, yet conventional implementations suffer from high operational labor, dependency on commercial fish feeds, energy-inefficient artificial lighting, and catastrophic ecosystem failures. This proposal presents **AquaFusion AI**, a novel, fully autonomous indoor aquaponics solution designed specifically for urban apartment/flat dwellers. The proposed system features: (1) **Computer Vision Fish Behavior Monitoring** using optical flow, trajectory tracking, and surface piping detection; (2) **Multi-modal Health Diagnostics** for both aquatic life and plants (NDVI analysis, leaf chlorosis detection, ORP/DO sensors); (3) **Feederless Bio-Integrated Fish Nutrition** utilizing overhead cultivated aquatic plant beds (Duckweed/Lemna minor and Wolffia) for continuous direct grazing; and (4) **Adaptive Overhead Green-Shifted Lighting** that extracts maximum electrical power conversion efficiency, dissipates only photosynthetic active radiation (PAR) frequencies, and dynamic height/intensity adaptation to achieve $> 2\times$ **crop yield** compared to standard windowsill/sunlight-restricted flat environments.

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1 Introduction & Problem Statement

Urban residential flat dwellers face significant space, light, and logistical constraints when attempting sustainable food production:

1. **Sunlight Deficit in Apartment Flats:** Windowsills and balconies provide intermittent, low-intensity natural sunlight, yielding sub-optimal photosynthetic rates and low crop growth.
2. **Energy Inefficiency of Full-Spectrum Grow Lights:** Broad-spectrum lights waste power across unabsorbed wavelengths, generating excessive heat and high electricity bills in confined flat living spaces.
3. **Commercial Feeder Dependence & Maintenance Burden:** Automated mechanical feeders clog, require frequent replenishment, and introduce excessive protein waste leading to toxic ammonia spikes.
4. **Ecosystem Fragility:** Sudden changes in dissolved oxygen (DO), pH, or ammonia (NH_3) can cause silent fish mass mortality before manual detection.

AquaFusion AI solves these challenges by combining AI edge computer vision, closed-loop aquatic plant feeding, and adaptive PAR lighting designed specifically for indoor urban agriculture.

2 Computer Vision Fish Behavior Monitoring

To eliminate manual inspection, an underwater RGB-D camera connected to an edge microcontroller (Raspberry Pi 5 / Jetson Orin Nano) continuously analyzes fish swimming kinetics.

2.1 Behavioral Metrics and Trajectories

The system tracks three principal behavioral indicators:

- **Swarming & Spatial Distribution:** Normal fish disperse homogeneously across the water volume. Aggregation near the surface indicates hypoxia ($\text{DO} < 4.0 \text{ mg/L}$), while bottom crowding indicates thermal stress or toxic ammonia spikes ($\text{NH}_3 > 0.05 \text{ mg/L}$).
- **Velocity Vector Profiling (Optical Flow):** Using Lucas-Kanade optical flow, normal cruising speed $\vec{v}_f$ is calculated. Lethargy ($\|\vec{v}_f\| \rightarrow 0$) or hyperactive darting ($\|\vec{v}_f\| \gg v_{\text{baseline}}$) triggers immediate alert protocols.
- **Surfacing / Piping Detection:** Frequent vertical movement toward the surface air-water interface is detected using YOLOV8-nano object tracking, signaling emergency oxygen depletion.

3 Investigating Fish & Plant Health

3.1 Fish Health Diagnostics

Fish health is evaluated dynamically via non-invasive multimodal sensing:

- **Symptom Detection:** Image segmentation models detect body lesions, white spots (*Ichthyophthirius multifiliis*), fin rot, and abnormal spinal curvature.
- **Water Quality Correlation:** Real-time chemical telemetry—Oxidation-Reduction Potential (ORP), Dissolved Oxygen (DO), pH, Electrical Conductivity (EC), Ammonia (NH_3), Nitrite (NO_2^-), and Nitrate (NO_3^-)—provides predictive root-cause diagnosis.

3.2 Plant Health Diagnostics

Plant foliage health is evaluated using top-down camera arrays:

- **Foliar Chlorosis & Deficiencies:** Computer vision monitors leaf color gradients. Yellowing older leaves indicate Nitrogen (N) deficiency, interveinal chlorosis indicates Iron (Fe) or Magnesium (Mg) deficiency, and purple tinting indicates Phosphorus (P) deficiency.
- **Growth Rate Estimation:** 3D depth sensors measure canopy volume expansion over 24-hour cycles.

4 Feederless Aquatic Plant-Based Feeding System

Traditional fish feeders are replaced by an integrated, self-sustaining aquatic plant foraging bed:

- **Live Bio-Feeder Bed:** High-protein fast-growing aquatic macrophytes (*Lemna minor* / Duckweed and *Wolffia globosa*) are cultivated in a shallow sub-chamber directly above or accessible to the fish tank.
- **Direct Grazing & Bio-Ramp:** Omnivorous and herbivorous fish (e.g., Tilapia, Koi, Gourami) continuously feed on fresh biomass enriched with plant-synthesized vitamins and protein (35 – 45% crude protein content).
- **Ammonia Recycling Loop:** Fish waste directly fertilizes the forage crop bed, removing toxic ammonia while producing fresh forage without human intervention or mechanical feeder jams.

5 Adaptive Overhead Green-Shifted PAR Lighting Mechanism

5.1 Resolving Flat-Dweller Sunlight Limitations (More Than 2x Yield)

In dense apartment flats, natural light intensity is typically limited to 50 – 150 $\mu\text{mol}/\text{m}^2/\text{s}$ for less than 4 hours daily. **AquaFusion AI** delivers optimal Photosynthetically Active

Radiation (PAR) continuously.

5.2 Narrow-Spectrum PAR Frequency Optimization & Heat Dissipation

- **Selective Wavelength Emission:** Plants utilize specific wavelengths (450nm Blue, 525 – 550nm Green penetration, 660nm Deep Red). Rather than burning energy across unabsorbed spectra (which causes excessive thermal radiation), AquaFusion AI uses high-efficiency solid-state LEDs emitting *only* exact photosynthetic frequencies.
- **Green Light Canopy Penetration:** Deep canopy leaves under shade are penetrated efficiently by tuned green spectrum wavelengths (525 – 550nm), enabling lower-tier leaves to perform photosynthesis at full capacity.
- **Power Extraction Efficiency:** By filtering out non-photosynthetic thermal infrared and UV waste, power conversion efficiency increases by $> 60\%$, dissipating minimal heat into flat living quarters.

5.3 Adaptive Height & Growth Motorized Tracking

- To maintain optimum Photosynthetic Photon Flux Density (PPFD) without burning growing tips, the overhead lighting array is mounted on motorized linear actuators.
- As the computer vision system detects canopy growth (Δh), the light bed automatically adjusts its height $h_{\text{light}}(t) = h_{\text{canopy}}(t) + d_{\text{optimal}}$, optimizing photon capture efficiency throughout the plant life cycle.

6 Expected Results & Metrics

Table 1: System Performance Comparisons

Metric	Standard Flat Sunlight	AquaFusion AI System
Daily PAR Yield ($\text{mol}/\text{m}^2/\text{day}$)	4.2 – 6.5	15.0 – 22.0 ($> 2\times$ Growth)
Feeder Maintenance	Refill every 3-5 days	Self-sustaining (0 Days)
Power Conversion Loss	High (Broad spectrum heat)	Minimal (Tuned PAR only)
Fish Survival Rate	75 – 80%	$> 98\%$

7 Conclusion

AquaFusion AI redefines urban indoor aquaponics. By integrating computer vision fish behavior tracking, automated plant/fish disease diagnostics, a feederless aquatic plant grazing mechanism, and adaptive green-penetration PAR lighting, flat dwellers can achieve more than double the crop yield of traditional natural-light farming while eliminating daily maintenance and minimizing power consumption.